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Thalamic Reticular Nucleus GABAergic Neurons: Structure, Connectivity, Markers, and Heterogeneity

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Background

TRN GABAergic neurons form the main inhibitory shell around the excitatory thalamus and act as a key gate for thalamic output. They help suppress irrelevant sensory and motor signals, shape thalamocortical activity across wake and sleep, and are central to sleep spindle generation. (5 sources)

In rodents, the thalamus is built around a largely glutamatergic relay “core” that is wrapped by a shell of GABAergic neurons corresponding mainly to the thalamic reticular nucleus (TRN). This organization makes the TRN the major inhibitory modulator of thalamic relay nuclei, with sector-specific reciprocal links to different thalamic domains. (Burnet-Merlin et al., 2026) (O'Reilly et al., 2020) In this role, TRN neurons regulate the excitability of thalamic nuclei, control the flow of information sent onward to cortex, and help set the dynamics of thalamocortical communication across active wake, quiet wake, and sleep. (Burnet-Merlin et al., 2026) (Crunelli et al., 2018)

Functionally, TRN GABAergic neurons provide a prominent and widespread source of tonic inhibition to thalamic relay nuclei. This inhibition is thought to suppress unimportant sensory input and unnecessary motor output, while transient reductions in TRN-mediated inhibition can support attention, arousal, and efficient sensorimotor transmission by disinhibiting relay circuits. (Brown et al., 2015) More broadly, TRN neurons are strongly linked to thalamic rhythm control: they are a key source for sleep spindle initiation during non-REM sleep, and during wakefulness they continue to regulate thalamocortical neuron activity rather than becoming silent. (Brown et al., 2015)

From a developmental systems view, the TRN is also important because maturation of its inhibitory function appears closely tied to the emergence of mature thalamocortical oscillations. Reviews of thalamic development argue that relatively weak early inhibition may permit amplification of immature thalamocortical activity, whereas later TRN maturation supports adult-like slow waves, delta activity, and sleep spindles. (Burnet-Merlin et al., 2026) (Murata et al., 2018)

Evidence

(Burnet-Merlin et al., 2026)

Implication of the thalamus in sleep alterations observed in Alzheimer's disease

"In rodents, the core of the thalamus is almost entirely composed of glutamatergic excitatory neurons. (O'Reilly et al., 2020) These glutamatergic nuclei are surrounded by a shell of GABAergic neurons, corresponding essentially to the thalamic reticular nucleus (TRN). The TRN is organized in sectors that are specifically and reciprocally connected to the different thalamic nuclei. Because the TRN contains only inhibitory neurons, it is considered the main modulator of the core nuclei's excitability. (Cox et al., 1997)(Murata et al., 2018) For instance, the TRN controls the flow of information delivered to the cortex and regulates the dynamics of the thalamocortical dialogue during the different vigilance states (active, quiet wake, and sleep). (Crunelli et al., 2018)"

"humans, inhibitory neurons can be found in most core thalamic nuclei, but the TRN remains the main source of inhibition to the core nuclei."

(O'Reilly et al., 2020)

Rodent somatosensory thalamocortical circuitry: neurons, synapses, and connectivity.

"As our understanding of the thalamocortical system deepens, the questions we face become more complex. Their investigation requires the adoption of novel experimental approaches complemented with increasingly sophisticated computational modeling. In this review, we take stock of current data and knowledge about the circuitry of the somatosensory thalamocortical loop in rodents, discussing common principles across modalities and species whenever appropriate. We review the different levels of organization, including the cells, synapses, neuroanatomy, and network connectivity. We provide a complete overview of this system that should be accessible for newcomers to this field while nevertheless being comprehensive enough to serve as a reference for seasoned neuroscientists and computational modelers studying the thalamocortical system. We further highlight key gaps in data and knowledge that constitute pressing targets for future experimental work. Filling these gaps would provide invaluable information for systematically unveiling how this system supports behavioral and cognitive processes."

(Crunelli et al., 2018)

Dual function of thalamic low-vigilance state oscillations: rhythm-regulation and plasticity

"During inattentive wakefulness and non-rapid eye movement (NREM) sleep, the neocortex and thalamus cooperatively engage in rhythmic activities that are exquisitely reflected in the electroencephalogram as distinctive rhythms spanning a range of frequencies from <1 Hz slow waves to 13 Hz alpha waves. In the thalamus, these diverse activities emerge through the interaction of cell-intrinsic mechanisms and local and long-range synaptic inputs. One crucial feature, however, unifies thalamic oscillations of different frequencies: repetitive burst firing driven by voltage-dependent Ca2+ spikes. Recent evidence reveals that thalamic Ca2+ spikes are inextricably linked to global somatodendritic Ca2+ transients and are essential for several forms of thalamic plasticity. Thus, we propose herein that alongside their rhythm-regulation function, thalamic oscillations of low-vigilance states have a plasticity function that, through modifications of synaptic strength and cellular excitability in local neuronal assemblies, can shape ongoing oscillations during inattention and NREM sleep and may potentially reconfigure thalamic networks for faithful information processing during attentive wakefulness."

(Brown et al., 2015)

Turning a Negative into a Positive: Ascending GABAergic Control of Cortical Activation and Arousal

"The most prominent, widespread, and well-known GABAergic input arises from the thalamic reticular nucleus (TRN)"

".In general, all of these GABAergic inputs maintain a high rate of tonic inhibition in the thalamic relay nuclei, which is likely important in suppressing unimportant information and unnecessary motor activity. However, in situations requiring high attention and responses to important situations, this tonic input is transiently suppressed allowing enhanced arousal/attention (disinhibition of higher-order nuclei) and sensorimotor transmission (disinhibition of first-order sensory and motor nuclei)"

".Thalamic reticular nucleus GABAergic neurons initiate sleep spindles in non-REM sleep. In addition, however, during wakefulness, they tonically regulate the activity of thalamocortical neurons"

".Several groups of GABAergic neurons in the thalamic reticular nucleus, ZI, and basal ganglia output nuclei control the activity of thalamic relay nuclei, suppressing unimportant sensory information and unnecessary motor activity."

(Murata et al., 2018)

Thalamic inhibitory circuits and network activity development

"Inhibitory circuits in thalamus and cortex shape the major activity patterns observed by electroencephalogram (EEG) in the adult brain. Their delayed maturation and circuit integration, relative to excitatory neurons, suggest inhibitory neuronal development could be responsible for the onset of mature thalamocortical activity. Indeed, the immature brain lacks many inhibition-dependent activity patterns, such as slow-waves, delta oscillations and sleep-spindles, and instead expresses other unique oscillatory activities in multiple species including humans. Thalamus contributes significantly to the generation of these early oscillations. Compared to the

abundance of studies on the development of inhibition in cortex, however, the maturation of thalamic inhibition is poorly understood. Here we review developmental changes in the neuronal and circuit properties of the thalamic relay and its interconnected inhibitory thalamic reticular nucleus (TRN) both in vitro and in vivo, and discuss their potential contribution to early network activity and its maturation. While much is unknown, we argue that weak inhibitory function in the developing thalamus allows for amplification of thalamocortical activity that supports the generation of early oscillations. The available evidence suggests that the developmental acquisition of critical thalamic oscillations such as slow-waves and sleep-spindles is driven by maturation of the TRN. Further studies to elucidate thalamic GABAergic circuit formation in relation to thalamocortical network function would help us better understand normal as well as pathological brain development."

Structure and anatomical location

The TRN is a thin, shell-like sheet of GABAergic neurons that wraps the anterior and lateral thalamus, with smaller dorsal and ventral extensions. Its position between thalamus and cortex is structural as well as functional: most thalamocortical and corticothalamic fibers pass through it and give off collaterals there. (9 sources)

Anatomically, the thalamic reticular nucleus is a thin lamellar or sheet-like nucleus that surrounds much of the dorsal thalamus, especially its anterior and lateral surfaces, with partial extension onto dorsal and ventral aspects. It lies between the outer medullary plate of the thalamus and the internal capsule, and is separated from adjacent thalamic relay nuclei by the external lamina. Across descriptions, it is consistently presented as a shell of neurons wrapped around the main thalamic mass rather than as a relay nucleus embedded within it. (Andres et al., 2011) (Pazo et al., 2013) (Biesbroek et al., 2023) (Gao et al., 2024) (Campbell et al., 2024)

Cellularly, the TRN is unusual because it is composed entirely, or essentially entirely, of GABAergic neurons. This makes it the main inhibitory shell around the otherwise largely excitatory dorsal thalamus. TRN neurons do not project directly to cortex; instead, their axons project broadly back to thalamic nuclei, with some reports also noting projections to mesencephalic tegmentum. (Andres et al., 2011) (Biesbroek et al., 2023) (Campbell et al., 2024) (Pazo et al., 2013)

Its placement is especially important because most axons running between cortex and thalamus pass through the TRN and innervate it via collateral branches. Both thalamocortical and layer VI corticothalamic fibers send excitatory collaterals into TRN, placing TRN neurons in a strategic position to sample and regulate traffic in both directions. In addition to these dominant thalamic and cortical inputs, TRN also receives afferents from forebrain and brainstem systems, including cholinergic, noradrenergic, and serotonergic sources, as well as basal ganglia-related inputs to at least some sectors. (Andres et al., 2011) (Villalobos et al., 2022) (Pazo et al., 2013) (Biesbroek et al., 2023)

The internal organization of TRN is topographic and sectorial. Reviews and circuit studies describe limbic, sensory, and motor sectors, with one limbic sector, five sensory sectors, and one motor sector proposed in rodents; the motor sector is placed rostrally. Within sensory TRN, connectivity can also be arranged across the thickness of the nucleus. For example, in rodent somatosensory TRN, neurons can be grouped into three tiers from the inner thalamic border to the outer border, each receiving input from different somatosensory relay nuclei in a roughly reciprocal pattern. (Villalobos et al., 2022) (Lam et al., 2011) (Pazo et al., 2013)

Recent work further suggests that TRN structure is not uniform across its width. In mouse, molecularly defined populations at opposite ends of a transcriptomic gradient map onto distinct anatomical arrangements described as core versus shell-like organization, and other summaries describe related “core/inner” and “edge” neurons with different soma size and dendritic extent. These features are better treated as candidate structural subtypes than as a single settled anatomical scheme, but they support the idea that the thin TRN sheet contains meaningful micro-organization across both sectors and local depth. (Flores et al., 2026) (Li et al., 2020)

Evidence

(Andres et al., 2011)

Functional Anatomy of Non-REM Sleep

"The thalamic reticular nucleus is a relatively thin sheet of neurons that surrounds the anterior, lateral and, to some extent, ventral surfaces of the dorsal thalamus, and it is entirely composed of GABAergic cells. Because of its anatomical position the thalamic reticular nucleus is traversed by virtually all axons connecting the dorsal thalamus with the neocortex, giving the nucleus its reticulated appearance and name; it also receives collateral branches from corticothalamic fibers originating in cortical layer VI (Figures 3 and 5). Thalamic reticular nucleus neurons do not project directly to the cortex although they widely project to the relay thalamic nuclei"

". Simultaneous intracellular recordings in reticular, thalamocortical, and cortical neurons (Steriade and Deschenes, 1988) indicated that sleep spindle generation was produced by a chain of events with three main consecutive steps: (1) repetitive spike-bursts at spindle frequency generated by GABAergic thalamic reticular nucleus neurons; (2) transfer of this activity to relay thalamocortical nuclei producing rhythmic (at spindle frequency) inhibitory postsynaptic potentials (IPSPs) in thalamocortical neurons; and (3) transfer to the cerebral cortex of postinhibitory rebound spike-bursts of the thalamocortical cells producing excitatory postsynaptic potentials (EPSPs) in cortical cells at spindle frequency. The repetitive spike-bursts of the reticular nucleus neurons are Na + action potentials generated by low-threshold Ca + spikes at spindle frequency (Bal et al., 1993). Since the reticular nucleus projection is GABAergic, the reticular nucleus activities are transmitted to the thalamocortical relay nuclei generating rhythmic IPSPs in these cells."

(Pazo et al., 2013)

Electrophysiologic study of globus pallidus projections to the thalamic reticular nucleus.

"The thalamic reticular nucleus (TRN) is a thin neuronal layer abutting the anterior and lateral aspect of the thalamus, partially extending to its dorsal and ventral parts (Huguenard and McCormick, 2007;Pinault, 2004). In addition to electrical coupling, the TRN neurons release the inhibitory neurotransmitter, GABA (Crabtree, 1999;Landisman et al., 2002;Steriade, 2005). Corticothalamic and thalamocortical fibers pass through the TRN giving off collaterals that innervate the reticular cells (Guillery et al., 1998;Pinault and Deschênes, 1998;Yu et al., 2009) and in turn, these reticular neurons project inhibitory axons to the thalamic nuclei (Huguenard and McCormick, 2007;Kolmac and Mitrofanis, 1997). Contrasting these reciprocal connections, there is no evidence of direct projections of the TRN to the cerebral cortex (Gandia et al., 1993;Jones, 1975;Haber et al., 1985).

Thalamocortical, corticothalamic and reticulothalamic projections are organized topographically (Cornwall et al., 1990;Crabtree and Killackey, 1989;Guillery et al., 1998;Lam and Sherman, 2011;Raos and Savaki, 1995;Shosaku et al., 1984;Shosaku and Sumitomo, 1983) resulting in auditory, visual, somatosensory and motor sectors identifiable within the TRN. In addition to thalamic and cortical afferents, the reticular cells receive afferents from the forebrain and various brainstem centers (Asanuma and Porter, 1990;Cornwall et al., 1990;Jones, 1975;Kolmac and Mitrofanis, 1998;McCormick, 1992;McAlonan et al., 2006;Sherman and Guillery, 1996)."

(Biesbroek et al., 2023)

When the central integrator disintegrates: A review of the role of the thalamus in cognition and dementia

"e reticular nucleus is a relatively thin nucleus wrapped around nearly the entire anterior and lateral surface of the thalamus.A thin layer of white matter called the external lamina separates the reticular nucleus from the other thalamic nuclei.The reticular nucleus is unique in the sense that it is believed to have no efferent projections to the cortex and only provides GABAergic output to other thalamic nuclei and to the mesencephalic tegmentum. 19,(Edlow et al., 2012)Its main input comes from other thalamic nuclei, striatum, basal forebrain, and brain stem nuclei including the ARAS.The reticular nucleus has a regulatory and "gating" function for activity in nearly all the other thalamic nuclei and, together with the intralaminar nuclei, modulates ARAS activation of thalamocortical networks. (Edlow et al., 2012)Thus, the reticular nucleus is critical for maintaining and regulating consciousness and attention"

". Interposed between the thalamus and the cortex, the reticular nucleus intercepts and regulates communication between these two structures (section 4.4). 54,57Because the reticular nucleus receives unidirectional projections from the cortex and has bidirectional connections with the thalamus, it is in a central position to modulate the flow of information between the two areas.Through this open-loop connection, the circuit is postulated to facilitate lateral inhibition, which may allow reticular nucleus cells to modulate thalamocortical cells in such a way as to amplify relevant stimuli to cortical areas while suppressing irrelevant stimuli. 61,62The reticular nucleus therefore represents the gate through which sensory information from other thalamic nuclei is relayed to the cortex.This gatekeeping role of the circuitry of the reticular nucleus depends on its physiological state.Whether the reticular nucleus is active depends largely on cholinergic and monoaminergic inputs from the brain stem that regulate the membrane potential of neurons of the reticular nucleus. 57By fine tuning the excitability of its neurons, the reticular nucleus allows for the functional flexibility it needs to selectively attend to relevant stimuli in a constantly changing environment.This explains its critical role in maintaining and regulation consciousness and attention."

(Gao et al., 2024)

Recent advances in neural mechanism of general anesthesia induced unconsciousness: insights from optogenetics and chemogenetics

"The thalamic reticular nucleus (TRN) is a thin lamellar layer of cells between the outer medullary plate of the thalamus and the internal capsule that surrounds the anterior and lateral aspects of the main body of the thalamus, consists of a shell of GABAergic neurons, receives inputs from cortical and subcortical regions, has crucial roles in sensory processing, arousal and cognition (Pinault, 2004;(McAlonan et al., 2006)(Halassa et al., 2014)(Deurveilher et al., 2011)"

".However, Liu Y et al. found that optogenetic and chemogenetic activation of the thalamic reticular nucleus GABAergic neurons shortened the emergence time under propofol anesthesia without altering the induction time. On the contrary, optogenetic and chemogenetic inhibition of the thalamic reticular nucleus GABAergic neurons prolonged the emergence time. They showed that activation of anterior thalamic reticular nucleus GABAergic neurons facilitates wakefulness from propofol anesthesia (Gui et al., 2021)."

(Campbell et al., 2024)

Development of reciprocal connections between the dorsal lateral geniculate nucleus and the thalamic reticular nucleus

"The thalamic reticular nucleus (TRN) is a shell-like structure that surrounds the dorsal and lateral aspects of the thalamus (Halassa et al., 2019)[2](Pinault, 2004)(Mitrofanis et al., 1993)(Guillery et al., 2002).Comprised entirely of GABAergic neurons, the TRN serves as an important nexus for thalamocortical (TC) and corticothalamic (CT) communication (Mitrofanis et al., 1993)(Houser et al., 1980)(Crick, 1984)(Steriade, 2005)(McCormick et al., 2014).This nucleus receives excitatory input from thalamocortical and corticothalamic axon collaterals, which in turn serve to activate GABAergic inhibitory feedback to many thalamic nuclei.This pattern of connectivity is arranged in a sectorial manner that allows for both modality-specific and network-wide interactions (Fig. 1)"

".Feedback projections from visual TRN to dLGN began to innervate dLGN as early as postnatal day 2 with weak inhibitory responses emerging during week 1. During week 2–4, inhibitory responses continued to grow larger, showing synaptic depression during repetitive stimulation. During this time TRN inhibition started to suppress TC spiking, having its greatest impact by week 4–6"

".Using a mutant mouse that lacks retinofugal projections revealed that the absence of retinal input led to an acceleration of TRN innervation of dLGN but had little impact on the development of feedforward projections from dLGN to TRN."

(Villalobos et al., 2022)

Elevation of GABA levels in the globus pallidus disinhibits the thalamic reticular nucleus and desynchronized cortical beta oscillations

"The reticular thalamic nucleus (RTn) articulates oscillations in large neuronal assemblies (Crick, 1984)(Halassa et al., 2016). GABAergic neurons in the RTn receive collaterals from thalamocortical (TC) and corticothalamic (CT) fibers and these connections modulate information transfer between thalamic nuclei and the cerebral cortex (Guillery et al., 2003)(Pinault, 2004). The RTn also receives inputs from noradrenergic, serotonergic, and cholinergic clusters in the brainstem (Asanuma, 1992)(Ben-Ari et al., 1976) (Cropper et al., 1984)(Hallanger et al., 1987)(Paré et al., 1988). These afferents control electrophysiological properties and regulate all TC activity. In this way, the RTn in combination with its afferents induces oscillations associated with functions such as attention, sleep, and consciousness (Clemente-Perez et al., 2017)(Crick, 1984)(Steriade, 2005)"

".The topographic organization of projections results in the formation of functional sectors in the RTn: one limbic sector, five sensory sectors, and one motor sector (Crabtree, 1992)(Lam et al., 2011). In rodents, the motor sector is in the rostral region of the RTn (Gonzalo-Ruiz et al., 1995)(Pinault et al., 1998)(Pinault, 2004) and receives GABAergic projections from the globus pallidus (GP; external GP in primates) (Asanuma et al., 1990)(Cornwall et al., 2004)(Gandia et al., 1993)(Hazrati et al., 1991)(Kayahara et al., 1998)."

(Lam et al., 2011)

Functional Organization of the Thalamic Input to the Thalamic Reticular Nucleus

"Most axons connecting the thalamus and cortex in both directions pass through the thalamic reticular nucleus (TRN), a thin layer of GABAergic cells adjacent to the thalamus, and innervate neurons there. The TRN, therefore, is in a strategic location to regulate thalamocortical communication. We recorded neurons of the somatosensory region of the TRN in a thalamocortical slice preparation and studied the spatial organization of their thalamic input using laser scanning photostimulation. We show that the thalamoreticular pathway is organized topographically for most neurons. The somatosensory region of the TRN can be organized into three tiers. From the inner (thalamoreticular) border to the outer, in a manner roughly reciprocal to the reticulothalamic pathway, each of these tiers receives its input from one of the somatosensory relays of the thalamus—the posterior medial, ventroposterior medial, and ventroposterior lateral nuclei. What is surprising is that approximately a quarter of the recorded neurons received input from multiple thalamic regions usually located in different nuclei. These neurons distribute evenly throughout the thickness of the TRN. Our results, therefore, suggest that there exist a subpopulation of TRN neurons that receive convergent inputs from multiple thalamic sources and engage in more complex patterns of inhibition of relay cells. We propose these neurons enable the TRN to act as an externally driven "searchlight" that integrates cortical and subcortical inputs and then inhibits or disinhibits specific thalamic relay cells, so that appropriate information can get through the thalamus to the cortex."

(Flores et al., 2026)

Thalamic reticular nucleus in the pathophysiology of schizophrenia

"At the cellular level, the TRN is composed of GABAergic inhibitory neurons"

"glutamate decarboxylase 67 and parvalbumin (PV), which is expressed in approximately 70% of these cells (Klimczak et al., 2024). These neurons exhibit distinct properties, including a specific topographic distribution. The TRN is primarily composed of two types of cells: core (inner neurons) and edge neurons, which have distinct structural and functional characteristics (Lam et al., 2011)(Li et al., 2020)(Martinez-Garcia et al., 2020). Core neurons are located mainly in the central region of the TRN and are distinguished by their larger somas and more extensive dendritic trees, allowing for greater integration of synaptic inputs. Additionally, these cells almost exclusively express secreted phosphoprotein 1 gene (Li et al., 2020)."

(Li et al., 2020)

Distinct subnetworks of the thalamic reticular nucleus

"The thalamic reticular nucleus (TRN), the major source of thalamic inhibition, regulates thalamocortical interactions that are critical for sensory processing, attention and cognition1–5. TRN dysfunction has been linked to sensory abnormality, attention deficit and sleep disturbance across multiple neurodevelopmental disorders6–9. However, little is known about the organizational principles that underlie its divergent functions. Here we performed an integrative study linking single-cell molecular and electrophysiological features of the mouse TRN to connectivity and systems-level function. We found that cellular heterogeneity in the TRN is characterized by a transcriptomic gradient of two negatively correlated gene-expression profiles, each containing hundreds of genes. Neurons in the extremes of this transcriptomic gradient express mutually exclusive markers, exhibit core or shell-like anatomical structure and have distinct electrophysiological properties. The two TRN subpopulations make differential connections with the functionally distinct first-order and higher-order thalamic nuclei to form molecularly defined TRN–thalamus subnetworks. Selective perturbation of the two subnetworks in vivo revealed their differential role in regulating sleep. In sum, our study provides a comprehensive atlas of TRN neurons at single-cell resolution and links molecularly defined subnetworks to the functional organization of thalamocortical circuits. A study integrating single-cell RNA-sequencing and electrophysiology data shows that in mouse, the cellular repertoire of the thalamic reticular nucleus is characterized by a transcriptomic gradient defined at its extremes by mutually exclusive expression of Spp1 and Ecel1, providing insights into the organizational principles underlying the divergent functions of this brain region."

Developmental origin and maturation

TRN GABAergic neurons come from inhibitory progenitors in the embryonic prethalamus, not from the excitatory thalamic progenitor zone. Their maturation is prolonged after birth: TRN-thalamic inhibition appears early but strengthens over weeks, and modulatory inputs such as cholinergic afferents are added in a sector-specific time course. (7 sources)

Developmentally, TRN neurons arise from GABAergic progenitors in the embryonic prethalamic compartment rather than from the excitatory thalamic lineage. In the developmental prosomeric scheme cited here, inhibitory progenitors including those that generate TRN are placed in p3/prethalamus, whereas excitatory thalamic neurons arise in p2. These GABAergic progenitors express **Ascl1** and enter a differentiation program that depends on **Dlx1**, **Dlx2**, and **Arx**, placing TRN neurons firmly within the prethalamic inhibitory lineage. (Sellers et al., 2014) (Sellers et al._1, 2014) (Jager et al., 2016)

Recent work further refines this developmental program by identifying an **Ascl1** → **Isl1** regulatory cascade that promotes TRN neuron fate while repressing alternative non-TRN prethalamic fates. In that study, this cascade was also required for proper formation of two major axonal systems in the region: thalamo-cortical projections and prethalamo-thalamic projections. Loss of this program disrupted prethalamo-thalamic axons and could secondarily disturb thalamo-cortical pathfinding within the thalamus, while forced **Isl1** expression rescued key defects in Ascl1-deficient TRN neurons in vitro. This suggests that TRN specification is tied closely to broader assembly of local prethalamic-thalamic and thalamocortical circuitry. (Nguyen et al., 2024)

Postnatally, TRN inhibitory output matures over an extended window. In the visual system, feedback projections from visual TRN to dLGN begin to innervate dLGN by about postnatal day 2, with weak inhibitory responses present in the first week. These responses increase through weeks 2-4, show synaptic depression during repetitive stimulation, and become increasingly able to suppress thalamocortical spiking, with the strongest impact reported by weeks 4-6. Retinal input is not required for the initial emergence of the dLGN → TRN feedforward projection, but loss of retinofugal input accelerates TRN innervation of dLGN, implying that peripheral sensory input can tune the timing of TRN feedback maturation rather than simply trigger it. (Campbell et al., 2024)

Neuromodulatory innervation also develops in a staged and sector-biased way. Cholinergic input to TRN

progresses roughly from ventral to dorsal sectors: nonvisual sensory sectors are innervated during week 1, the visual sector during week 2, and by week 3 cholinergic fibers are denser across the nucleus. Functional cholinergic responses follow a similar sequence, with weak nicotinic excitation appearing first and later muscarinic inhibition emerging during weeks 3-4, so that more mature responses become biphasic. This indicates that TRN maturation is not just intrinsic to the GABAergic neuron lineage but also depends on the timed addition of state-related modulatory control. (Sokhadze et al., 2018)

Maturation is also visible at the level of marker expression. **Parvalbumin** is widely used as a TRN neuron marker in mature tissue, and in rat postnatal development its immunoreactive pattern changes across age and across different TRN regions, with regional variation in the timing of maturation and an overall decline in staining intensity with age. Because this evidence is based on immunoreactivity, it supports **PV protein** maturation rather than directly addressing transcript levels. (Malinova et al., 2013) Combined with the developmental physiology above, these data support a view in which TRN neurons are specified embryonically as prethalamic GABAergic cells, then acquire mature inhibitory efficacy, neuromodulatory responses, and mature marker patterns over the first postnatal weeks.

Evidence

(Sellers et al., 2014)

Transcriptional control of GABAergic neuronal subtype identity in the thalamus

> *"The ability to modulate the flow of information is supported by a second abundant neuronal type, the inhibitory GABA neurons of the reticular nucleus of the thalamus (TRN). TRN neurons are defined by the expression of the parvalbumin gene (Biasi et al., 1994), their fast spiking action and specific connectivity, which is strictly confined within the thalamus. While TRN neurons are a near homogeneous population"*

> *"The TRN neurons derive from GABAergic progenitors in the embryonic prethalamic (pTh) compartment (Cobos et al., 2005)."*

(Sellers et al._1, 2014)

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Tectal-derived interneurons contribute to phasic and tonic inhibition in the visual thalamus

> *"Within the thalamus, GABAergic drive depends on two functionally distinct neuronal populations: intrinsic interneurons, and neurons of the reticular thalamic nucleus (RTN) (Houser et al., 1980)(Hunt et al., 1991)"*

> *".Inhibitory progenitors, including those of the RTN, are found in p3; whilst excitatory neurogenesis takes place in p2 (refs 13,14)"*

> *".GABAergic progenitors in p3, including those of the RTN, express Ascl1 and progress through a differentiation programme that depends on Dlx1, Dlx2 and Arx (refs 11,16-21)."*

(Nguyen et al., 2024)

Regulation of neuronal fate specification and connectivity of the thalamic reticular nucleus by the Ascl1-Isl1 transcriptional cascade

> *"The thalamic reticular nucleus (TRN) is an anatomical and functional hub that modulates the flow of information between the cerebral cortex and thalamus, and its dysfunction has been linked to sensory disturbance and multiple behavioral disorders"*

> *".Here, we showed that the regulatory cascade of the transcription factors Ascl1 and Isl1 promotes the fate of TRN neurons and concomitantly represses the fate of non-TRN prethalamic neurons. Furthermore, we found that this cascade is necessary for the correct development of the two main axonal connections, thalamo-cortical projections and prethalamo-thalamic projections. Notably, the disruption of prethalamo-thalamic axons can cause the pathfinding defects of thalamo-cortical axons in the thalamus. Finally, forced Isl1 expression can rescue disruption of cell fate specification and prethalamo-thalamic projections in in vitro primary*

cultures of Ascl1-deficient TRN neurons, indicating that Isl1 is an essential mediator of Ascl1 function in TRN development."

(Campbell et al., 2024)

Development of reciprocal connections between the dorsal lateral geniculate nucleus and the thalamic reticular nucleus

"The thalamic reticular nucleus (TRN) is a shell-like structure that surrounds the dorsal and lateral aspects of the thalamus (Halassa et al., 2019)[2](Pinault, 2004)(Mitrofanis et al., 1993)(Guillery et al., 2002).Comprised entirely of GABAergic neurons, the TRN serves as an important nexus for thalamocortical (TC) and corticothalamic (CT) communication (Mitrofanis et al., 1993)(Houser et al., 1980)(Crick, 1984)(Steriade, 2005)(McCormick et al., 2014).This nucleus receives excitatory input from thalamocortical and corticothalamic axon collaterals, which in turn serve to activate GABAergic inhibitory feedback to many thalamic nuclei.This pattern of connectivity is arranged in a sectorial manner that allows for both modality-specific and network-wide interactions (Fig. 1)"

".Feedback projections from visual TRN to dLGN began to innervate dLGN as early as postnatal day 2 with weak inhibitory responses emerging during week 1. During week 2–4, inhibitory responses continued to grow larger, showing synaptic depression during repetitive stimulation. During this time TRN inhibition started to suppress TC spiking, having its greatest impact by week 4–6"

".Using a mutant mouse that lacks retinofugal projections revealed that the absence of retinal input led to an acceleration of TRN innervation of dLGN but had little impact on the development of feedforward projections from dLGN to TRN."

(Sokhadze et al., 2018)

Postnatal development of cholinergic input to the thalamic reticular nucleus of the mouse

"The thalamic reticular nucleus (TRN), a shell-like structure comprised of GABAergic neurons, gates signal transmission between thalamus and cortex. While TRN is innervated by axon collaterals of thalamocortical and corticothalamic neurons, other ascending projections modulate activity during different behavioral states such as attention, arousal, and sleep-wake cycles. One of the largest arise from cholinergic neurons of the basal forebrain and brainstem"

". Cholinergic innervation of TRN follows a ventral-to-dorsal progression, with nonvisual sensory sectors receiving input during week 1, and the visual sector during week 2. By week 3, the density of cholinergic fibers increases throughout TRN and forms a reticular profile. Functional patterns of connectivity between cholinergic fibers and TRN neurons progress in a similar manner, with weak excitatory nicotinic responses appearing in nonvisual sectors near the end of week 1. By week 2, excitatory responses become more prevalent and arise in the visual sector. Between weeks 3–4, inhibitory muscarinic responses emerge, and responses become biphasic, exhibiting a fast excitatory, and a long-lasting inhibitory component."

(Malinova et al., 2013)

Parvalbumin Immunoreactivity Changes during the Maturation of the Thalamic Reticular Nucleus of the Rat

"The thalamic reticular nucleus is a thin layer of GABAergic cells located between the external medullary lamina and the internal capsule surrounding the rostolateral surface of the thalamus. It has functionally distinct afferent and efferent connections with thalamic nuclei, the neocortex, the basal forebrain and the brainstem. Parvalbumin is a calcium-binding protein, which is regarded to be a marker for GABAergic neurons. The thalamic reticular neurons are GABAergic, and parvalbumin is always colocalized with GABA in these cells. We have demonstrated the parvalbumin immunoreactivity in the thalamic reticular nucleus at different stages of postnatal development of rats, as well at 1-year-old rats. It was established that the maturation of immunoposive patterns varies in different parts of the nucleus. The intensity of immunostaining decreases with age."

Connectivity

TRN GABAergic neurons sit in a classic reciprocal loop with thalamic relay cells and cortex: they receive excitatory collaterals from thalamocortical and corticothalamic axons, then send inhibitory output back to thalamus. In addition, they are connected to one another by local inhibitory and electrical coupling, and some TRN sectors also receive strong extrathalamic inhibition, including from the external globus pallidus. (11 sources)

TRN connectivity is organized around a reciprocal thalamic gate. TRN neurons are excited by glutamatergic collaterals from thalamic relay neurons and by corticothalamic fibers from layer VI, and in turn they project inhibitory axons back onto thalamic relay nuclei in an orderly topographic way. This arrangement follows directly from their position in the path of thalamocortical radiations and has been described across sensory systems and species. (Graybiel et al., 1983) (Paz et al., 2021) In somatosensory cat TRN, response latencies and receptive field properties were taken to support activation by collaterals of ventral posterior thalamocortical relay axons, and single filled TRN neurons sent thin axons with diffuse branches across much of ventral posterior and posterior thalamus. (Yen et al., 1985)

The output side of this loop is inhibitory and can be anatomically specific at the level of relay-cell dendrites.

In visual thalamus, GABAergic “F terminals” in dLGN arise from both TRN neurons and local interneurons, placing TRN as a major nonretinal inhibitory input to retinogeniculate transmission. (Hammer et al., 2014) (Hammer et al._1, 2014) Ultrastructural work further suggests that most nRT-to-thalamus terminals make single synapses and preferentially contact thin distal dendrites of relay cells, consistent with broad but fine-grained inhibitory control. (Wanaverbecq et al., 2008)

Connectivity within the TRN itself is also important. Anatomical and physiological studies support mutual inhibition among TRN neurons, mediated mainly by GABA_A receptors and in some cases also GABA_B receptors. Filled perigeniculate/TRN neurons often show intrareticular axon collaterals, and local activation can suppress neighboring TRN cells, a motif proposed to sharpen thalamic inhibition by producing local center-surround-like effects. (Yen et al., 1985) (Sanchez-Vives et al., 1997) (Long et al., 2004) (Ulrich et al., 1995) That said, the strength and prevalence of chemical intra-TRN synapses may vary by subtype and context rather than being uniform across the whole nucleus.

TRN neurons are also linked by electrical synapses. In mouse and rat, gap-junction-coupled clusters form at least two patterns: elongated clusters that run within functional modules in the plane of TRN and have axons that selectively inhibit local relay-cell groups, and clusters that span the thickness of TRN and project to both first-order and higher-order relay nuclei. Dye coupling is reduced but not eliminated in connexin36 knockout mice, supporting connexin36-dependent electrical coupling plus possible additional coupling mechanisms. (Lee et al., 2014)

Finally, TRN connectivity includes notable extrathalamic inhibitory control. A recent study reports that all recorded TRN neurons received inhibition from Npas1-expressing neurons in the external globus pallidus, while only a subset also received intra-TRN inhibition; in that dataset, intra-TRN inhibition was biased from somatostatin-expressing onto parvalbumin-expressing TRN neurons, suggesting subtype-specific local wiring layered onto a more global pallidal inhibitory input. (Cho et al., 2025) Together, these data support a connectivity scheme with several nested motifs: reciprocal relay loops, local chemical and electrical TRN networks, and sector- or subtype-specific extrathalamic inputs that can coordinate activity across the TRN.

Evidence

(Graybiel et al., 1983)

Somatostatin-like immunoreactivity characterizes neurons of the nucleus reticularis thalami in the cat and monkey

> *"The nucleus reticularis of the thalamus lies like a shell"*
>
> *"around the lateral and anterior sides of the main thalamic mass so that its neurons stand in the path of the thalamic radiations. This remarkable physical arrangement has a direct counterpart in the connections that neurons of the nucleus reticularis make: they are innervated by collaterals of the thalamocortical and corticothalamic fibers sweeping through the nucleus, and they project back upon the thalamus in an orderly and apparently topo-"*
>
> *"the inhibition exerted by the nucleus reticularis operates by means of y-aminobutyric acid (GABA) because they could demonstrate, in neurons of the nucleus, high immunoreactivity to the synthetic enzyme of the GABA mechanism, glutamic acid decarboxylase (GAD). We report here that at least the majority of neurons in the nucleus reticular-is of the cat are also characterized by immunoreactivity to antisera prepared against somatostatin, a neuropeptide itself reported to have mainly inhibitory effects on neural and hormonal function (see Iversen et al., 1978a, b;Reichlin, 1980)"*
>
> *".In each brain, intense somatostatin-like immunoreactivity was found in neurons of the nucleus reticularis thalami. The immunoreactivity was localized to the perinuclear region of the neurons in both species. Somatostatin-like immunoreactivity appeared in only a sharply restricted subset of other zones thought to have an embryologic origin in the ventral thalamus"*
>
> *"The neurons of the nucleus reticularis also showed moderate acetylcholinesterase activity but were negative in tests of immunoreactivity to antisera raised against avian pancreatic polypeptide and Met-enkephalin. We conclude that the presence of somatostatin-like immunoreactivity may be a defining characteristic of the nucleus reticularis thalami and suggest that the functional activity of the neurons of the nucleus reticularis may involve not only a GABA mechanism but also the action of somatostatin or a related neuropeptide."*

(Paz et al., 2021)

The role of GABA neurotransmitter in the human central nervous system, physiology, and pathophysiology

"The thalamocortical pathway consists of GABAergic neurons that project from the thalamic reticular nucleus (nRT) toward the ventral basalis (VB), generating an inhibitory "loop" for a self-modulation of the nRT. GABAergic neurons of the nRT receive the glutamatergic stimulation from the VB and the cortico-thalamic fibers that are projected from the layer six of the cerebral cortex. In this "loop context," GABA A receptors are in glutamatergic thalamocortical and cortico-thalamic neurons, as well as in the GABAergic neurons themselves (McCormick et al., 1997) ."

(Yen et al., 1985)

The morphology of physiologically identified GABAergic neurons in the somatic sensory part of the thalamic reticular nucleus in the cat

"Neurons with somatic sensory receptive fields were examined electrophysiologically in the thalamic reticular nucleus of the cat. All cells had receptive fields much larger than those of neurons in the ventral posterior nucleus and were driven by less readily defined somesthetic stimuli. Response latencies to peripheral or medial lemniscal stimulation were, on average, longer than in the ventral posterior nucleus and suggested activation of the reticular nucleus cells by collaterals of thalamocortical relay cell axons arising in the ventral posterior nucleus. When injected intracellularly with horseradish peroxidase, reticular nucleus cells displayed thin axons with intrareticular collaterals and diffuse branches through much of the ventral posterior and posterior thalamic nuclei. Dendrites ended in processes resembling synaptic terminals. Electron microscopic immunocytochemistry of the same part of the reticular nucleus revealed processes immunoreactive for glutamic acid decarboxylase and identifiable as both collateral axon terminals and presynaptic dendrites of GABAergic reticular nucleus cells. These synaptically linked reticular nucleus cells and, in addition, immunoreactive somata and presynaptic dendrites received synapses from at least three varieties of nonimmunoreactive profiles."

(Hammer et al., 2014)

Nuclei-specific differences in nerve terminal distribution, morphology, and development in mouse visual thalamus

"The gain of retinogeniculate signal transmission is modulated by nonretinal inputs to dLGN. These nonretinal inputs arise from visual cortex, pretectum, brainstem, thalamic reticular nuclei, and local dLGN interneurons, and they far outnumber the more powerful retinal inputs (Sherman et al., 2002)(Sherman, 2004)"

".One set of nerve terminals contain flattened, oval shaped synaptic vesicles, a hallmark feature of inhibitory GABAergic terminals, and are therefore called F terminals (Sherman, 2004)(Lund et al., 1972)(Famiglietti, 1970). In dLGN, GABAergic terminals arise from the thalamic reticular nucleus and local inhibitory interneurons [7,18]19]."

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(Wanaverbecq et al., 2008)

Contrasting the Functional Properties of GABAergic Axon Terminals with Single and Multiple Synapses in the Thalamus

"one input from the reticular thalamic nucleus (nRT)"

"The vast majority of nRT-thalamic terminals formed single synapses per postsynaptic target and innervated thin distal dendrites of relay cells."

(Sanchez-Vives et al., 1997)

Inhibitory Interactions between Perigeniculate GABAergic Neurons

"Perigeniculate neurons form an interactive sheet of cells that inhibit one another as well as thalamocortical neurons in the dorsal lateral geniculate nucleus (LGNd)"

".These IPSPs displayed an average reversal potential of −77 mV and were blocked by application of bicuculline methiodide or picrotoxin, indicating that they are mediated by GABAA receptors. In the presence of GABAA receptor blockade, the activation of PGN neurons with glutamate could result in slow IPSPs that were mediated by GABAB receptors in a subset (40%) of cells. Similarly, application of specific agonists muscimol and baclofen demonstrated that PGN neurons possess both functional GABAAand GABAB receptors. Examination of the axon arbors of biocytin-filled PGN neurons often revealed the presence of beaded axon collaterals within the PGN, suggesting that this may be an anatomical substrate for PGN to PGN inhibition. Functionally, activation of

inhibition between PGN neurons could result in a shortening or a complete abolition of the low threshold Ca2+ spike or an inhibition of tonic discharge. We suggest that the mutual inhibition between PGN neurons forms a mechanism by which the excitability of these cells is tightly controlled. The activation of a point within the PGN may result in the inhibition of neighboring PGN neurons. This may be reflected in the LGNd as a center of inhibition surrounded by an annulus of disinhibition, thus forming a "center-surround" mechanism for thalamic function"

".Previous studies of the GABAergic neurons of the thalamic reticular nucleus have demonstrated that they exhibit functional GABA A receptors that when activated result in increases in membrane Cl conductances (McCormick and Prince, 1986;Bal et al., 1995b;Ulrich and Huguenard, 1995). In addition, recent studies by Ulrich and Huguenard (1996) have demonstrated that activation of the local neuropil in the rodent thalamic reticular nucleus with electrical stimulation can activate GABA A and occasionally GABA B receptor-mediated IPSPs. Presumably these IPSPs arise from the activation of axon collaterals from other thalamic reticular neurons; however, because these investigators used electrical stimulation to evoke these events, it is also possible Local application of acetazolamide (200 M in micropipette) blocks the depolarizing response and reveals an underlying hyperpolarization that has distinct rapid and slow phases. B, The rapid phase reverses at approximately 83mV and is blocked by picrotoxin. After the application of picrotoxin, the slow phase remains and exhibits a reversal potential of 95mV. C, In the presence of picrotoxin, slowly increasing the application of GABA results in progressive increases in the amplitude of the slow hyperpolarization. that they originate from axons of extrathalamic sources, such as the substantia nigra reticulata or the basal forebrain and other forebrain structures (Jourdain et al., 1989;Pare et al., 1990;Asanuma, 1994). Indeed, recently Pinault et al. (1995Pinault et al. (, 1997)) suggested, on the basis of morphological studies, that axonal synaptic interactions between thalamic reticular cells in the adult rat are rare. Our results clearly demonstrate, with local activation of perigeniculate GABAergic cells with glutamate, that these neurons exert potent inhibitory on one another, typically through the activation of GABA A receptors but also occasionally through the activation of GABA B receptors."

(Long et al., 2004)

Small Clusters of Electrically Coupled Neurons Generate Synchronous Rhythms in the Thalamic Reticular Nucleus

"The GABAergic neurons that comprise the thalamic reticular nucleus (TRN) surround and inhibit the neurons in the thalamic relay nuclei (Steriade et al., 1997;(Crabtree, 1999). TRN neurons often participate in widespread rhythmic, synchronized activity that encompasses other thalamic nuclei and the neocortex (Steriade et al., 1987)(Krosigk et al., 1993)"

".Axons originating outside the TRN, such as glutamatergic afferents from the neocortex and thalamic relay nuclei, may help to synchronize the firing of TRN cells (Destexhe et al., 1998)(Bal et al., 2000)(Blumenfeld et al., 2000). Synaptic connectivity within the TRN is also likely to play an important role. Anatomical (Deschenes et al., 1985;Yen et al., 1985;De Biasi et al., 1988;Liu and Jones, 1999) and physiological (Ulrich and Huguenard, 1996;Sanchez-Vives et al., 1997;Shu and McCormick, 2002) evidence suggest that TRN neurons form mutually inhibitory GABAergic synapses."

(Ulrich et al., 1995)

Purinergic inhibition of GABA and glutamate release in the thalamus: implications for thalamic network activity.

"Adenosine is a CNS depressant with both pre- and postsynaptic actions. Presynaptically, adenosine decreases neurotransmitter release in the hippocampus but only at excitatory terminals. In the thalamus, however, we show that, in addition to its actions at excitatory synapses, adenosine strongly suppresses monosynaptic inhibitory currents both in relay cells of the thalamic ventrobasal complex (VB) and in inhibitory neurons of the nucleus reticularis thalami (nRt). A concomitant increase in transmission failures and results coefficient of variation analysis are both consistent with a presynaptic mechanism. Pharmacological manipulations support an A1 receptor-mediated process. Slow thalamic oscillations induced in vitro by extracellular stimulation and recorded with extracellular multiunit electrodes in VB and nRt are dampened by adenosine without affecting their periodicity. We conclude that adenosine can presynaptically down-regulate inhibitory postsynaptic responses in thalamus and exert robust antioscillatory effects, likely by synergistic depression of both excitatory and inhibitory neurotransmitter release."

(Lee et al., 2014)

Two Functionally Distinct Networks of Gap Junction-Coupled Inhibitory Neurons in the Thalamic Reticular Nucleus

"GABAergic neurons in the thalamic reticular nucleus (TRN) of mice and rats form two types of GJ-coupled clusters with distinctive patterns and axonal projections. Most clusters are elongated narrowly along functional modules within the plane of the TRN, with axons that selectively inhibit local groups of relay neurons. However, some coupled clusters have neurons arrayed across the thickness of the TRN and target their axons to both first- and higher-order relay nuclei. Dye coupling was reduced, but not abolished, among cells of connexin36 knock-out mice. Our results suggest that GJs form two distinct types of inhibitory networks that correlate activity either within or across functional modules of the thalamus."

(Cho et al., 2025)

A Dual Inhibitory Network in the Thalamic Reticular Nucleus Delineated by Pallidal and Intra-Reticular Inhibition

"Long described as an inhibitory "guardian of the gateway," the thalamic reticular nucleus (TRN) shapes which thalamic signals reach the cortex during attention, arousal, and sensory processing"

". All recorded TRN neurons received inhibition from the GPe, whereas a subset also received intra-TRN inhibition. Intra-TRN inhibition arose predominantly from somatostatin-expressing onto parvalbumin-expressing TRN neurons (SOM PV)revealing subtype-specific connectivity recruited by thalamic excitation to form a feedforward motif"

". Sparse, dendrite-targeting intra-TRN connections from parvalbumin- and somatostatin-expressing neurons provide structured, feedforward inhibition recruited by thalamocortical (TC) inputs. In parallel, robust and widespread inhibition from Npas1-expressing neurons of the external globus pallidus (GPe) supplies a strong external drive that can coordinate activity across the TRN. Together, these complementary motifs define a dual inhibitory architecture that may shape thalamocortical gating and highlight cellular entry points for therapeutic intervention."

Function

TRN GABAergic neurons act as a state-dependent gate on thalamic output. They help switch thalamocortical circuits between external signal transmission, selective attention, sleep rhythms, and recovery from anesthesia. (5 sources)

At the systems level, TRN neurons are best understood as inhibitory controllers of thalamocortical traffic. Because they receive excitatory drive linked to both thalamic relay activity and cortical feedback, they can suppress or permit the passage of signals through thalamus depending on behavioral state, making them important for sensory processing, arousal, and cognition. (Gao et al., 2024) (Halassa et al., 2014)

A classic function of TRN is sleep rhythm control, especially sleep spindle generation. Electrophysiological work and later reviews describe TRN as a pacemaker for spindles, and during sleep onset increased thalamic-reticular activity is proposed to inhibit thalamocortical relay neurons, in part via GABA_B-dependent inhibition, thereby reducing sensory transfer to cortex during non-REM sleep. (Alrousan et al., 2022)

TRN function is not uniform across all of its sectors. Ensemble recording and circuit-mapping work supports distinct subnetworks: limbic-projecting TRN neurons track arousal, whereas sensory-projecting TRN neurons participate in spindles, become more synchronized during slow-wave sleep, and are suppressed during attention-demanding states. In that framework, TRN can separately regulate externally driven sensory processing and internally generated brain activity. (Gao et al., 2024) (Halassa et al., 2014)

In awake behavior, TRN also contributes to selective attention. In visual TRN of awake monkeys, neurons show strong and fast visual responses, and the transient response increases when attention is directed to the visual stimulus, supporting the long-standing idea that TRN helps shape an attentional “searchlight” over thalamic maps by modulating inhibition onto relay nuclei. (Gao et al., 2024) (McAlonan et al., 2006)

Recent causal studies further suggest that TRN function depends on anatomical sector and task domain. In one reported example, activating anterior TRN GABAergic neurons shortened emergence from propofol anesthesia, whereas inhibiting them prolonged emergence, arguing that at least this anterior sector can promote waking from an anesthetized state rather than simply enforcing sensory blockade. (Gao et al., 2024)

Beyond sensory gating and arousal, TRN GABAergic neurons have also been linked to social cognition. A recent study reports that parvalbumin interneurons in sensory TRN are necessary and sufficient for social memory in mice, and places them in a cortico-reticular-thalamic-parafascicular pathway involved in social memory engrams. This extends TRN function beyond classic sensory and sleep roles into behaviorally specific cognitive functions. (Gao, 2024)

Evidence

(Gao et al., 2024)

Recent advances in neural mechanism of general anesthesia induced unconsciousness: insights from optogenetics and chemogenetics

"The thalamic reticular nucleus (TRN) is a thin lamellar layer of cells between the outer medullary plate of the thalamus and the internal capsule that surrounds the anterior and lateral aspects of the main body of the thalamus, consists of a shell of GABAergic neurons, receives inputs from cortical and subcortical regions, has crucial roles in sensory processing, arousal and cognition (Pinault, 2004;(McAlonan et al., 2006)(Halassa et al., 2014)(Deurveilher et al., 2011)"

".However, Liu Y et al. found that optogenetic and chemogenetic activation of the thalamic reticular nucleus GABAergic neurons shortened the emergence time under propofol anesthesia without altering the induction time. On the contrary, optogenetic and chemogenetic inhibition of the thalamic reticular nucleus GABAergic neurons prolonged the emergence time. They showed that activation of anterior thalamic reticular nucleus GABAergic neurons facilitates wakefulness from propofol anesthesia (Gui et al., 2021)."

(Halassa et al., 2014)

State-dependent architecture of thalamic reticular sub-networks

"Behavioral state is known to influence interactions between thalamus and cortex, which are important for sensation, action and cognition. The thalamic reticular nucleus (TRN) is hypothesized to regulate thalamo-cortical transmission, but the underlying functional architecture of this process and its state-dependence are unknown. By combining the first TRN ensemble recording with psychophysics and connectivity-based optogenetic tagging, we find that the TRN is composed of distinct sub-networks. While activity of limbic-projecting TRN neurons correlates with arousal, sensory-projecting neurons participate in spindles and show elevated synchrony by slow waves during sleep. Conversely, sensory-projecting neurons are suppressed by attentional states, demonstrating common microcircuit mechanisms of sensory processing in sleep and attention. Bidirectional manipulation of attentional performance was achieved through optogenetic manipulation of these TRN sub-networks. Our findings provide evidence for differential regulation of thalamic inhibition across brain states, suggesting that the TRN separately controls external stimulus processing and internally-generated computations, a basic determinant of cognitive function."

(Alrousan et al., 2022)

Early Life Sleep Deprivation and Brain Development: Insights From Human and Animal Studies

"Data reported in various electrophysiological studies have suggested that the thalamic reticular nucleus seems to function as a pacemaker of sleep spindles generation (Steriade et al., 1993). The thalamic reticular nucleus serves as the sensory and internal signal operating at the gateway toward the cerebral cortex brain region and consists of exclusively the GABAergic neurons"

".During the sleep initiation process, reduction in wake-promoting cells is believed to lead to reduction in activity of thalamocortical relay and the enhancement of thalamic-reticular cellular activity. Excitation of thalamic-reticular is responsible for further inhibiting thalamocortical relay by activation of the inhibitory postsynaptic GABA-B receptors (Steriade et al., 1993). Consequently, cerebral cortex sensory signals at the cortical gate are blocked within the thalamus, depriving the cerebral cortex of receiving external information, as evident from high voltage and slow brain waves during NREM sleep, which is an indication of cortical inactivity (Stenberg and Porkka-Heiskanen, 1990;Kalia, 2006;Datta, 2010)."

(McAlonan et al., 2006)

Attentional Modulation of Thalamic Reticular Neurons

"The major pathway for visual information reaching cerebral cortex is through the lateral geniculate nucleus (LGN) of the thalamus. Acting on this vital relay is another thalamic nucleus, the thalamic reticular nucleus (TRN). This nucleus receives topographically organized collaterals from both thalamus and cortex and sends similarly organized projections back to thalamus. The inputs to the TRN are excitatory, but the output back to the thalamic relay is inhibitory, providing an ideal organization for modulating visual activity during early processing. This functional architecture led Crick in 1984 to hypothesize that TRN serves to direct a searchlight of attention to different regions of the topographic map; however, despite the substantial influence of this hypothesis, the activity of TRN neurons has never been determined during an attention task. We have determined the nature of the response of visual TRN neurons in awake monkeys, and the modulation of that response as the monkeys shifted attention between visual and auditory stimuli. Visual TRN neurons had a strong (194 spikes/s) and fast (25 ms latency) transient increase of activity to spots of light falling in their receptive fields, as well as high background firing rate (45 spikes/s). When attention shifted to the spots of light, the amplitude of the transient visual response typically increased, whereas other neuronal response characteristics remained unchanged. Thus, as predicted previously, TRN activity is modified by shifts of visual attention, and these attentional changes could influence visual processing in LGN via the inhibitory connections back to the thalamus."

(Gao, 2024)

Thalamic cells and pathway for social memory processing and storage

"parvalbumin interneurons in the sensory thalamic reticular nucleus are necessary and sufficient for regulating social memory in mice, identify a novel cortico-reticular thalamic-parafascicular pathway for social cognition, and highlight an essential role of GABAergic inhibitory neurons in social memory engrams."

Neurotransmitters, receptors, and neuropeptides

TRN neurons are classically GABAergic and are defined in part by strong expression of the GABA-synthetic enzymes GAD65 and especially GAD67. Beyond GABA, they show distinctive receptor complements for serotonin, glutamate, and GABA itself, and in at least cat and monkey many TRN neurons also contain somatostatin-like peptide. (10 sources)

The core transmitter phenotype of TRN neurons is GABAergic. Classic anatomical work described the TRN as densely packed with neurons containing GABA, and immunostaining for glutamate decarboxylase (GAD) was used as early evidence that TRN inhibition is mediated by GABA. (Delfs et al., 1996) (Graybiel et al., 1983) At the transcript level, single-cell in situ hybridization in rat detected both **Gad1/GAD67** and **Gad2/GAD65** mRNAs in TRN neurons, with **GAD67 mRNA** showing a stronger regulation than **GAD65 mRNA** after dopaminergic lesions, especially in lateral and ventral TRN. This supports the idea that both GABA-synthetic isoforms are present, but that their expression can be differentially regulated by circuit state. (Delfs et al., 1996)

On the receptor side, TRN neurons both release GABA and are themselves shaped by GABA receptor signaling. A notable feature is that, unlike many thalamocortical relay neurons, TRN neurons appear not to express the **GABA_A** receptor **α4** and **δ** subunits that support a strong extrasynaptic tonic current in relay cells; correspondingly, no tonic GABA_A current was observed in NRT neurons in the cited slice work. (Lee et al._1, 2014) (Cope et al., 2005) This fits with earlier developmental in situ data indicating that NRT does not show the same postnatal switch in GABA_A receptor subunit expression that is seen in ventrobasal relay nuclei. (Lee et al._1, 2014) (Laurie et al., 1992)

TRN neurons are also strong targets of serotonin. Review and physiological evidence cited here indicates that TRN GABAergic neurons express both **5-HT_{1A}** and **5-HT_{2A}** receptors, with **5-HT_{1A} receptor mRNA** detected by in situ hybridization in some thalamic nuclei and with functional evidence for both receptor types at presynaptic TRN terminals. In ventrobasal recordings, serotonin reduced miniature IPSC frequency arising from TRN input, and this effect depended on **5-HT_{1A}** and **5-HT_{2A}** receptors, supporting a presynaptic serotonergic control of GABA release from TRN axons. (Inserra et al., 2021) (Pompeiano et al., 1992) (Goitia et al., 2015) Inserra et al. also note that TRN receives serotonergic input from dorsal raphe and noradrenergic input from locus coeruleus, and that these modulatory systems help keep TRN in a slightly depolarized state. (Inserra et al., 2021)

Glutamate acts on TRN neurons through metabotropic as well as ionotropic mechanisms. In rat TRN slices, activation of **group I mGluRs** caused depolarization, whereas activation of **group II mGluRs** caused hyperpolarization, with both effects linked to changes in a linear K^+ conductance. This is notable because glutamatergic drive to TRN is usually thought of as excitatory; the group II mGluR effect provides a built-in route for glutamate to reduce TRN firing and thereby disinhibit relay nuclei. (Inserra et al., 2021) (Cox et al., 1999)

For neuropeptides, the strongest evidence in the supplied papers is for **somatostatin-like immunoreactivity** in TRN neurons. In cat and monkey, Graybiel and colleagues reported that at least the majority of TRN neurons showed somatostatin-like immunoreactivity, with labeling concentrated around the perinuclear region, and suggested that this may be a defining feature of the nucleus in those species. The same study reported that TRN neurons were negative for **Met-enkephalin** and **avian pancreatic polypeptide** immunoreactivity, while showing moderate acetylcholinesterase activity. (Graybiel et al., 1983) This peptide result should be interpreted carefully in light of newer mouse work: in current rodent circuit studies, **SOM** more often functions as a subtype marker for one TRN GABAergic population rather than as a claim that all TRN neurons are peptidergically equivalent. (Inserra et al., 2021) (Clemente-Perez et al., 2017)

Finally, recent work suggests transmitter-related specialization across TRN subtypes. In mouse, **PV^+** and **SOM^+** GABAergic TRN neurons differ in physiology and circuit role: PV^+ cells are described as rhythmogenic, while SOM^+ cells are linked more to limbic and higher-order thalamic circuits. These papers do not define distinct fast transmitter usage between the two groups—both are GABAergic—but they do support subtype-specific molecular programs layered onto the shared GABA phenotype. (Inserra et al., 2021) (Clemente-Perez et al., 2017)

Evidence

(Delfs et al., 1996)

Unilateral nigrostriatal lesions induce a bilateral increase in glutamate decarboxylase messenger RNA in the reticular thalamic nucleus.

"The reticular thalamic nucleus consists of densely packed neurons containing the neurotransmitter GABA. It surrounds the lateral border of the thalamus, has extensive reciprocal connections with thalamocortical neurons, and is thought to be involved in attentional processes. The reticular thalamic nucleus also receives direct and indirect inputs from the basal ganglia, suggesting that it may be involved in relaying motor information to the thalamus and cortex"

".Sections of the reticular thalamic nucleus were processed for in situ hybridization histochemistry at the single cell level with RNA probes for both isoforms of glutamate decarboxylase (M(r) 65,000: glutamate decarboxylase 65 and M(r) 67,000: glutamate decarboxylase 67), the rate limiting enzyme of GABA synthesis. Unilateral nigrostriatal dopaminergic lesions induced a topographically specific, bilateral increase in glutamate decarboxylase 67 messenger RNA in neurons of the lateral and ventral reticular thalamic nucleus. A much smaller increase in glutamate decarboxylase 65 messenger RNA was observed which was significant only ipsilateral to the lesion."

(Graybiel et al., 1983)

Somatostatin-like immunoreactivity characterizes neurons of the nucleus reticularis thalami in the cat and monkey

"The nucleus reticularis of the thalamus lies like a shell"

"around the lateral and anterior sides of the main thalamic mass so that its neurons stand in the path of the thalamic radiations. This remarkable physical arrangement has a direct counterpart in the connections that neurons of the nucleus reticularis make: they are innervated by collaterals of the thalamocortical and corticothalamic fibers sweeping through the nucleus, and they project back upon the thalamus in an orderly and apparently topo-"

"the inhibition exerted by the nucleus reticularis operates by means of y-aminobutyric acid (GABA) because they could demonstrate, in neurons of the nucleus, high immunoreactivity to the synthetic enzyme of the GABA mechanism, glutamic acid decarboxylase (GAD). We report here that at least the majority of neurons in the nucleus reticular-is of the cat are also characterized by immunoreactivity to antisera prepared against somatostatin, a neuropeptide itself reported to have mainly inhibitory effects on neural and hormonal function (see Iversen et al., 1978a, b;Reichlin, 1980)"

".In each brain, intense somatostatin-like immunoreactivity was found in neurons of the nucleus reticularis thalami. The immunoreactivity was localized to the perinuclear region of the neurons in both species. Somatostatin-like immunoreactivity appeared in only a sharply restricted subset of other zones thought to have an embryologic origin in the ventral thalamus"

"The neurons of the nucleus reticularis also showed moderate acetylcholinesterase activity but were negative in tests of immunoreactivity to antisera raised against avian pancreatic polypeptide and Met-enkephalin. We conclude that the presence of somatostatin-like immunoreactivity may be a defining characteristic of the nucleus reticularis thalami and suggest that the functional activity of the neurons of the nucleus reticularis may involve not only a GABA mechanism but also the action of somatostatin or a related neuropeptide."

(Lee et al._1, 2014)

The impact of tonic GABAA receptor-mediated inhibition on neuronal excitability varies across brain region and cell type

"While cells in the ventrobasal nuclei (VB) undergo a developmental switch postnatally in their expression of GABA A R subunits, neurons in the nucleus reticularis thalami (NRT) do not undergo such a switch (Laurie et al., 1992)"

".In contrast, no tonic current was observed in NRT cells, which do not express α4 or δ subunits (Cope et al., 2005)(Jia et al., 2005)."

(Cope et al., 2005)

GABAA Receptor-Mediated Tonic Inhibition in Thalamic Neurons

"Tonic GABAA receptor-mediated inhibition is typically generated byδ subunit-containing extrasynaptic receptors. Because the δ subunit is highly expressed in the thalamus, we tested whether thalamocortical (TC) neurons of the dorsal lateral geniculate nucleus (dLGN) and ventrobasal complex exhibit tonic inhibition. Focal application of gabazine (GBZ) (50 μm) revealed the presence of a 20 pA tonic current in 75 and 63% of TC neurons from both nuclei, respectively. No tonic current was observed in GABAergic neurons of the nucleus reticularis thalami (NRT). Bath application of 1μm GABA increased tonic current amplitude to 7ϕA in 100% of TC neurons, but it was still not observed in NRT neurons. In dLGN TC neurons, the tonic current was sensitive to low concentrations of the δ subunit-specific receptor agonists allotetrahydrodeoxycorticosterone (100 nm) and 4,5,6,7-tetrahydroisoxazolo[5,4-c]-pyridin-3-ol (THIP) (100 nm) but insensitive to the benzodiazepine flurazepam (5 μm). Bath application of low concentrations of GBZ (25–200 nm) preferentially blocked the tonic current, whereas phasic synaptic inhibition was primarily maintained. Under intracellular current-clamp conditions, the preferential block of the tonic current with GBZ led to a small depolarization and increase in input resistance. Using extracellular single-unit recordings, block of the tonic current caused the cessation of low-threshold burst firing and promoted tonic firing. Enhancement of the tonic current by THIP hyperpolarized TC neurons and promoted burst firing. Thus, tonic current in TC neurons generates an inhibitory tone. Its modulation contributes to the shift between different firing modes, promotes the

transition between different behavioral states, and predisposes to absence seizures."

(Laurie et al., 1992)

The distribution of thirteen GABAA receptor subunit mRNAs in the rat brain. III. Embryonic and postnatal development

"The embryonic and postnatal expression of 13 GABAA receptor subunit genes in the rat CNS was studied by in situ hybridization. Each transcript exhibited a unique regional and temporal developmental expression profile. For example, in both embryonic and early postnatal cortex and thalamus, expression of the alpha 2, alpha 3, alpha 5, and beta 3 mRNAs was pronounced. In particular, the alpha 5 gene expression underwent a prominent peak in early brain. Subsequently, the thalamocortical expression of these four genes substantially diminished and was superseded in the adult by the alpha 1, alpha 4, beta 2, and delta subunit mRNAs. Similarly, gamma 1 and gamma 3 gene expression also dropped markedly during development, their initial stronger expression being restricted to relatively few structures. In contrast, gamma 2 gene expression was widespread and mostly remained constant with increasing age. The medial septum and globus pallidus were regions expressing few subunits in both early postnatal and adult stages, allowing clear developmental combinatorial changes to be inferred (alpha 2/alpha 3 beta 2 gamma 2 to alpha 1 beta 2 gamma 2, alpha 2/alpha 3 beta 2 gamma 1 to alpha 1 beta 2 gamma 1/gamma 2, respectively). In contrast, cerebellar Purkinje cells exhibited no developmental switch, expressing only the alpha 1, beta 2, beta 3, and gamma 2 mRNAs from birth to adult. Certain GABAA transcripts were also detected in germinal zones (e.g., beta 1, beta 3, gamma 1) and in embryonic peripheral tissues such as dorsal root ganglia (e.g., alpha 2, alpha 3, beta 3, gamma 2) and intestine (gamma 3). Some parallels in regional and temporal CNS expression were noted (e.g., alpha 1 beta 2, alpha 2 beta 3, alpha 4/alpha 6 delta), whereas the alpha 5 and beta 1 regional mRNA expressions converted over time. The changes of GABAA receptor subunit gene expression suggest a molecular explanation for earlier observations on changing ligand binding affinities. Thus, the composition, and presumably properties, of embryonic/early postnatal rat GABAA receptors differs markedly from those expressed in the adult brain."

(Inserra et al., 2021)

Lysergic acid diethylamide differentially modulates the reticular thalamus, mediodorsal thalamus, and infralimbic prefrontal cortex: An in vivo electrophysiology study in male mice

"the reticular thalamus gates thalamocortical information flow via finely tuned inhibition of thalamocortical cells in the mediodorsal thalamus"

". two subpopulations with distinct intrinsic membrane excitability have been characterized by Clemente-Perez et al. (2017) in the mouse reticular thalamus, which are somatostatin (SOM+) and parvalbumin (PV+) GABAergic neurons, two biomarkers which can also be used to differentiate subpopulations of reticular thalamus GABAergic neurons in humans. While SOM+ neurons are linked to limbic circuitries, and project to higher order relay nuclei, PV+ are rhythmogenic and seem to project mostly to first-order relay nuclei (Clemente-Perez et al., 2017). However, more studies are necessary to better characterize these different populations of reticular thalamus neurons, correlating them with their physiological functions"

". Indeed, GABAergic neurons of the reticular thalamus express pre-and postsynaptic 5-HT 2A and 5-HT 1A receptors (Aznar et al., 2003)(Goitia et al., 2015)(Pompeiano et al., 1992)(Santana et al., 2011) and receive serotonergic projections from the DRN, the brain areas in which most of the serotonergic neurons are located (Rodríguez et al., 2011). The reticular thalamus receives also norepinephrinergic projections from the locus coeruleus (Asanuma, 1992), which together with the 5-HT/ dorsal raphe nucleus (DRN) input keeps the reticular thalamus in a slightly depolarized state."

(Pompeiano et al., 1992)

Distribution and cellular localization of mRNA coding for 5-HT1A receptor in the rat brain: correlation with receptor binding

"In order to localize the cells expressing 5-HT1A receptors in the rat brain, we used in situ hybridization histochemistry to visualize the distribution of the mRNA coding for 5-HT1A receptors. Oligonucleotides derived from different parts of the coding region of the rat 5-HT1A receptor gene were used as hybridization probes. 5-HT1A binding sites were visualized on consecutive sections by receptor autoradiography using 3H-8-hydroxy-2-(di-n-propylamino)tetralin as ligand. The highest levels of hybridization were observed in the dorsal raphe nucleus, septum, hippocampus, entorhinal cortex, and interpeduncular nucleus. Positive hybridization signals were also present in other areas, such as the olfactory bulb; cerebral cortex; some thalamic and hypothalamic nuclei; several nuclei of the brainstem, including all the remaining raphe nuclei, nucleus of the solitary tract, and nucleus of the spinal tract of the trigeminus; and the dorsal horn of the spinal cord. The distribution and abundance of 5-HT1A receptor mRNA in different rat brain areas generally correlate with those of the binding sites, suggesting that 5-HT1A receptors are predominantly somatodendritic receptors."

(Goitia et al., 2015)

MODULATION OF GABA RELEASE FROM THE THALAMIC RETICULAR NUCLEUS BY COCAINE AND CAFFEINE: ROLE OF SEROTONIN RECEPTORS

"Serotonin receptors are targets of drug therapies for a variety of neuropsychiatric and neurodegenerative disorders. Cocaine inhibits the re-uptake of serotonin (5-HT), dopamine, and noradrenaline, whereas caffeine blocks adenosine receptors and opens ryanodine receptors in the endoplasmic reticulum. We studied how 5-HT and adenosine affected spontaneous GABAergic

transmission from thalamic reticular nucleus. We combined whole-cell patch clamp recordings of miniature inhibitory post-synaptic currents (mIPSCs) in ventrobasal thalamic neurons during local (puff) application of 5-HT in wild type (WT) or knockout mice lacking 5-HT2A receptors (5-HT2A−/−). Inhibition of mIPSCs frequency by low (10 μM) and high (100 μM) 5-HT concentrations was observed in ventrobasal neurons from 5-HT2A−/− mice. In WT mice, only 100 μM 5-HT significantly reduced mIPSCs frequency. In 5-HT2A−/− mice, NAN-190, a specific 5-HT1A antagonist, prevented the 100 μM 5-HT inhibition while blocking H-currents that prolonged inhibition during post-puff periods. The inhibitory effects of 100 μM 5-HT were enhanced in cocaine binge-treated 5-HT2A−/− mice. Caffeine binge treatment did not affect 5-HT-mediated inhibition. Our findings suggest that both 5-HT1A and 5-HT2A receptors are present in pre-synaptic thalamic reticular nucleus terminals. Serotonergic-mediated inhibition of GABA release could underlie aberrant thalamocortical physiology described after repetitive consumption of cocaine."

(Cox et al., 1999)

Glutamate Inhibits Thalamic Reticular Neurons

"Activation of metabotropic glutamate receptors (mGluRs) can result in long-lasting modulation of neuronal excitability. Multiple mGluR subtypes are localized within the rat thalamic reticular nucleus (TRN), and we have examined the effects of activating these different receptor subtypes on the excitability of these neurons using an in vitro slice preparation. Typical of most mGluR-sensitive preparations, the general mGluR agonist, (±)-1-aminocyclopentane-trans-1,3-dicarboxylic acid (ACPD) produced a robust, long-lasting excitatory response. Surprisingly, ACPD produced a membrane hyperpolarization in some neurons. Using selective mGluR agonists, we found that activation of group II mGluRs produces the hyperpolarization, whereas the depolarization is mediated by group I mGluRs. While the polarity of the postsynaptic response (hyperpolarization vs depolarization) was dependent on the mGluR subtype activated, both actions appear to result from modification of a linear K+ conductance. The inhibitory action of Glutamate, via group II mGluRs, provides an avenue for a disinhibitory effect that could have interesting consequences upon a well-investigated, model neuronal circuit, turning its assumed functional role upside down."

(Clemente-Perez et al., 2017)

Distinct thalamic reticular cell types differentially modulate normal and pathological cortical rhythms

"SUMMARY Integrative brain functions depend on widely distributed, rhythmically coordinated computations. Through its long-ranging connections with cortex and most senses, the thalamus orchestrates the flow of cognitive and sensory information. Essential in this process, the nucleus reticularis thalami (nRT) gates different information streams through its extensive inhibition onto other thalamic nuclei; however, we lack an understanding of how different inhibitory neuron subpopulations in nRT function as gatekeepers. We dissociated the connectivity, physiology, and circuit functions of neurons within rodent nRT, based on parvalbumin (PV) and somatostatin (SOM) expression, and validated the existence of such populations in human nRT. We found that PV but not SOM cells are rhythmogenic, and that PV and SOM neurons are connected to and modulate distinct thalamocortical circuits. Notably, PV but not SOM neurons modulate somatosensory behavior and disrupt seizures. These results provide a conceptual framework for how nRT may gate incoming information to modulate brain-wide rhythms."

Markers and anatomical context

*In practice, TRN neurons are most often marked by their shared GABAergic identity plus strong enrichment for parvalbumin, but PV is not fully universal and is best treated as a common TRN marker in rodent anatomical context rather than a sole defining feature. Newer work also shows that somatostatin and parvalbumin can mark distinct TRN subpopulations, while transcriptomic studies identify mutually exclusive molecular extreme markers such as **Spp1** and **Ecel1**. (5 sources)*

- **GABAergic markers: VGAT, GAD1, GAD2**
- **Type of marker:** transcript/protein class markers for inhibitory neurons.
- **Anatomical context:** useful for identifying TRN neurons as part of the inhibitory shell around dorsal thalamus, but **not specific to TRN** because many other inhibitory populations also express them.
- In the supplied material, nearby PHb GABA neurons are described as coexpressing **Vgat, Gad1, and Gad2**, illustrating that these markers identify GABAergic fate but do not on their own separate TRN from neighboring inhibitory cell groups. (Weil et al., 2022)
- For TRN specifically, these markers are best used together with anatomical location or with more selective markers such as PV, SOM, or transcriptomic subtype markers. (Weil et al., 2022)
- **Parvalbumin (PV / Pvalb)**
- **Type of marker:** mainly used here as a **protein marker** by immunoreactivity; also commonly used as a **transcript/transgene marker** in modern mouse studies, though the supplied quotes here directly support

protein expression and general expression status.

- **Anatomical context:** a widely used marker for rodent TRN neurons; in rodents, PV^+ inhibitory neurons in thalamus are described as being located almost exclusively in the TRN, making PV especially useful for distinguishing TRN from most other thalamic territories. (Klimczak et al., 2024) (Malinova et al., 2013)
- Many TRN neurons express PV, and PV immunoreactivity has been followed across postnatal development in rat TRN. In that context, PV is always colocalized with GABA in TRN cells, and the staining pattern matures at different rates in different parts of the nucleus. (Malinova et al., 2013)
- A practical caveat is that PV should not be treated as a universal marker for every TRN neuron. More recent work explicitly contrasts **PV** with **SOM** as "classic markers of TRN neurons," implying partly distinct TRN populations rather than a single uniform PV^+ class. (Weil et al., 2022) (Li et al., 2020)
- **Somatostatin (SOM / Sst)**
- **Type of marker:** used in recent mouse work as a **transcript/transgene-defined subtype marker**; older peptide immunoreactivity studies also reported somatostatin-like labeling, but in current anatomical use SOM mainly marks a TRN subpopulation rather than the whole nucleus.
- **Anatomical context:** helps distinguish one TRN subtype from others, especially relative to PV^+ TRN neurons. It is also useful for separating TRN from neighboring inhibitory populations that lack both PV and SOM.
- Weil et al. state that PHb GABA neurons do **not** express parvalbumin or somatostatin, explicitly citing these as classic TRN markers. This makes **PV/SOM negativity** useful for ruling out TRN identity in adjacent inhibitory thalamic regions. (Weil et al., 2022) (Li et al., 2020)
- In the broader TRN literature summarized in the earlier sections, SOM is best viewed as a marker of a TRN subpopulation, not a pan-TRN marker. (Li et al., 2020)
- **Spp1 and Ecel1**
- **Type of marker: transcript markers** from single-cell transcriptomics.
- **Anatomical context:** these are not general inhibitory markers; instead, they label opposite ends of a molecular gradient within mouse TRN and therefore act as **subtype markers inside TRN**.
- Li et al. reported that the TRN contains a transcriptomic gradient with mutually exclusive markers at its extremes: **Spp1** and **Ecel1**. These molecular extremes also map onto different anatomical structure, described as more **core** versus **shell-like** organization, and onto different thalamic connectivity patterns. (Li et al., 2020)
- Thus, **Spp1** and **Ecel1** are useful when the goal is not simply to identify TRN, but to divide TRN neurons into molecularly and anatomically distinct subnetworks. (Li et al., 2020)
- **nNOS**
- **Type of marker:** protein marker in the cited context.
- **Anatomical context: not a TRN marker** in the supplied material; instead, it was previously shown to mark a subpopulation of **dLGN interneurons**.
- Weil et al. use absence of **nNOS** together with lack of PV and SOM to define PHb GABA neurons relative to nearby cell classes, and cite prior evidence that nNOS delineates a subclass of dLGN interneurons. This is useful anatomically because it helps separate TRN neurons from local inhibitory neurons in visual thalamus. (Weil et al., 2022) (Leist et al., 2016)

- **How these markers are best used together**
- For **pan-TRN identification**, the most reliable approach in the supplied literature is **anatomical location + GABAergic identity + strong PV enrichment**, while keeping in mind that some TRN neurons instead fall into SOM-associated or transcriptomically distinct classes. (Klimczak et al., 2024) (Malinova et al., 2013) (Li et al., 2020)
- For **distinguishing TRN from nearby inhibitory nuclei**, **PV and/or SOM positivity** can help, whereas **VGAT/GAD1/GAD2 alone** cannot. (Weil et al., 2022)
- For **distinguishing subtypes within TRN**, the most informative markers in the supplied papers are **PV vs SOM** at the population level and **Spp1 vs Ecel1** at the transcriptomic-gradient extremes. (Weil et al., 2022) (Li et al., 2020)

Evidence

(Weil et al., 2022)

Daily changes in light influence mood via inhibitory networks within the thalamic perihabenular nucleus

> *"PHb GABA neurons, which are defined by the coexpression of Vgat, Gad1, and Gad2, do not express parvalbumin or somatostatin, classic markers of TRN neurons (Li et al., 2020)(256823087), or nNOS, previously shown to delineate a subpopulation of dLGN interneurons (Leist et al., 2016)."*

(Klimczak et al., 2024)

Long-term effects of a double hit murine model for schizophrenia on parvalbumin expressing cells and plasticity-related molecules in the thalamic reticular nucleus and the habenula

> *"Fast-spiking parvalbumin (PV) expressing GABAergic neurons play an important role in brain physiology by providing inhibitory control of cortical and subcortical circuits (Hu et al., 2014). They are involved in cortical maturation through activity-dependent changes and the control of critical periods (Carulli et al., 2021)(Reichelt et al., 2019). These inhibitory neurons undergo a protracted period of physiological maturation and integration into cortical and thalamic circuitry during early postnatal life (Reh et al., 2020)"*

> *".In the thalamus, the TRN is the major source of GABAergic inhibition, and modulates the information between the cerebral cortex and the thalamic nuclei (Takata, 2019). Interestingly, many inhibitory neurons in this nucleus express PV and, in rodents, the thalamic PV+ inhibitory neurons are almost exclusively located in the TRN (Lein et al., 2007). The projections of these neurons contact with the rest of the thalamic nuclei, including the lateral posterior nucleus (LP), the homologous of the human pulvinar nucleus, and the mediodorsal nucleus (MD) (Pinault, 2004)."*

(Malinova et al., 2013)

Parvalbumin Immunoreactivity Changes during the Maturation of the Thalamic Reticular Nucleus of the Rat

> *"The thalamic reticular nucleus is a thin layer of GABAergic cells located between the external medullary lamina and the internal capsule surrounding the rostolateral surface of the thalamus. It has functionally distinct afferent and efferent connections with thalamic nuclei, the neocortex, the basal forebrain and the brainstem. Parvalbumin is a calcium-binding protein, which is regarded to be a marker for GABAergic neurons. The thalamic reticular neurons are GABAergic, and parvalbumin is always colocalized with GABA in these cells. We have demonstrated the parvalbumin immunoreactivity in the thalamic reticular nucleus at different stages of postnatal development of rats, as well at 1-year-old rats. It was established that the maturation of immunoposive patterns varies in different parts of the nucleus. The intensity of immunostaining decreases with age."*

(Li et al., 2020)

Distinct subnetworks of the thalamic reticular nucleus

> *"The thalamic reticular nucleus (TRN), the major source of thalamic inhibition, regulates thalamocortical interactions that are critical for sensory processing, attention and cognition1–5. TRN dysfunction has been linked to sensory abnormality, attention deficit and sleep disturbance across multiple neurodevelopmental disorders6–9. However, little is known about the organizational principles that underlie its divergent functions. Here we performed an integrative study linking single-cell molecular and electrophysiological features of the mouse TRN to connectivity and systems-level function. We found that cellular heterogeneity in the TRN is characterized by a transcriptomic gradient of two negatively correlated gene-expression profiles, each containing hundreds of genes. Neurons in the extremes of this transcriptomic gradient express mutually exclusive markers, exhibit core or shell-like anatomical structure and have distinct electrophysiological properties. The two TRN subpopulations make differential connections with the functionally distinct first-order and higher-order thalamic nuclei to form molecularly defined TRN–thalamus subnetworks.*

Selective perturbation of the two subnetworks in vivo revealed their differential role in regulating sleep. In sum, our study provides a comprehensive atlas of TRN neurons at single-cell resolution and links molecularly defined subnetworks to the functional organization of thalamocortical circuits. A study integrating single-cell RNA-sequencing and electrophysiology data shows that in mouse, the cellular repertoire of the thalamic reticular nucleus is characterized by a transcriptomic gradient defined at its extremes by mutually exclusive expression of Spp1 and Ecel1, providing insights into the organizational principles underlying the divergent functions of this brain region."

(Leist et al., 2016)

Two types of interneurons in the mouse lateral geniculate nucleus are characterized by different h-current density

"Although hyperpolarization-activated cyclic nucleotide-gated cation (HCN) channels and the corresponding h-current (Ih) have been shown to fundamentally shape the activity pattern in the thalamocortical network, little is known about their function in local circuit GABAergic interneurons (IN) of the dorsal part of the lateral geniculate nucleus (dLGN). By combining electrophysiological, molecular biological, immunohistochemical and cluster analysis, we characterized the properties of Ih and the expression profile of HCN channels in IN. Passive and active electrophysiological properties of IN differed. Two subclasses of IN were resolved by unsupervised cluster analysis. Small cells were characterized by depolarized resting membrane potentials (RMP), stronger anomalous rectification, higher firing frequency of faster action potentials (APs), appearance of rebound bursting, and higher Ih current density compared to the large IN. The depolarization exerted by sustained HCN channel activity facilitated neuronal firing. In addition to cyclic nucleotides, Ih in IN was modulated by PIP2 probably based on the abundant expression of the HCN3 isoform. Furthermore, only IN with larger cell diameters expressed neuronal nitric oxide synthase (nNOS). It is discussed that Ih in IN is modulated by neurotransmitters present in the thalamus and that the specific properties of Ih in these cells closely reflect their modulatory options."

Transcriptomics and molecular cell types

*Single-cell work in mouse suggests that TRN GABAergic neurons are not split into many fully separate classes in the simplest sense, but are arranged along a strong molecular gradient. The best-supported molecular extremes are **Spp1**-rich and **Ecel1**-rich populations, which also differ in anatomy, physiology, thalamic targets, and sleep-related function. (1 source)*

The main transcriptomic dataset in the supplied papers comes from an integrative single-cell study of mouse TRN that combined single-cell RNA-seq with electrophysiology and circuit analysis. In that study, TRN heterogeneity was described as a **transcriptomic gradient** rather than a set of sharply isolated clusters, with two negatively correlated gene-expression programs spanning the population and each containing hundreds of genes. Cells at the two ends of this gradient expressed **mutually exclusive transcript markers, Spp1 and Ecel1**, providing the clearest molecular anchors for TRN cell typing in the cited literature. (Li et al., 2020)

This transcriptomic organization was linked directly to other cell properties. The two molecular extremes did not only differ in RNA profile; they also showed different **electrophysiological features**, different **anatomical structure** described as more **core** versus **shell-like**, and different **connectivity** with thalamic nuclei. In particular, the study reported that these molecularly defined TRN populations make different connections with **first-order** versus **higher-order** thalamic nuclei, supporting the idea that transcriptomic identity in TRN is tied to specific thalamic subnetworks rather than being only a descriptive molecular label. (Li et al., 2020)

The same work also linked transcriptomic cell type to systems function in vivo. Selective perturbation of the two molecularly defined TRN-thalamus subnetworks produced different effects on **sleep regulation**, showing that the molecular gradient has behavioral relevance and is not just a correlational pattern. This is a notable result because it connects single-cell transcriptomics to anatomy, physiology, connectivity, and function in one framework. (Li et al., 2020)

For the user's marker-focused question, the most important transcript-level conclusion is therefore that **Spp1** and **Ecel1** function as **transcriptomic subtype markers within mouse TRN**, specifically for cells at opposite ends of the major molecular axis. In anatomical context, they are most useful for dividing TRN neurons into internal subpopulations, not for separating all TRN neurons from the rest of thalamus by themselves. The supplied quote does not provide a public accession number, so that cannot be added here from the provided source alone. (Li et al., 2020)

Overall, the available transcriptomic evidence in the supplied set supports a model in which TRN GABAergic

neurons share a common inhibitory identity but vary along a structured molecular continuum. At present, the best-characterized molecular cell types in these papers are the **Spp1-extreme** and **Ecel1-extreme** TRN neurons, with intermediate cells likely occupying positions between those poles rather than forming wholly unrelated categories. (Li et al., 2020)

Evidence

(Li et al., 2020)

Distinct subnetworks of the thalamic reticular nucleus

> *"The thalamic reticular nucleus (TRN), the major source of thalamic inhibition, regulates thalamocortical interactions that are critical for sensory processing, attention and cognition1–5. TRN dysfunction has been linked to sensory abnormality, attention deficit and sleep disturbance across multiple neurodevelopmental disorders6–9. However, little is known about the organizational principles that underlie its divergent functions. Here we performed an integrative study linking single-cell molecular and electrophysiological features of the mouse TRN to connectivity and systems-level function. We found that cellular heterogeneity in the TRN is characterized by a transcriptomic gradient of two negatively correlated gene-expression profiles, each containing hundreds of genes. Neurons in the extremes of this transcriptomic gradient express mutually exclusive markers, exhibit core or shell-like anatomical structure and have distinct electrophysiological properties. The two TRN subpopulations make differential connections with the functionally distinct first-order and higher-order thalamic nuclei to form molecularly defined TRN–thalamus subnetworks. Selective perturbation of the two subnetworks in vivo revealed their differential role in regulating sleep. In sum, our study provides a comprehensive atlas of TRN neurons at single-cell resolution and links molecularly defined subnetworks to the functional organization of thalamocortical circuits. A study integrating single-cell RNA-sequencing and electrophysiology data shows that in mouse, the cellular repertoire of the thalamic reticular nucleus is characterized by a transcriptomic gradient defined at its extremes by mutually exclusive expression of Spp1 and Ecel1, providing insights into the organizational principles underlying the divergent functions of this brain region."*

Heterogeneity and characterized subtypes

*TRN GABAergic neurons are heterogeneous, but the strongest evidence supports a major internal axis rather than many fully separate classes. The best-characterized subtypes are a central **Spp1\(^+\)** / core-like / first-order-projecting / burst-prone population and a border **Ecel1\(^+\)** / shell-like / higher-order-projecting / less burst-prone population, with additional heterogeneity in local input convergence and in PV-defined versus non-PV cells. (5 sources)*

The current picture is that TRN heterogeneity is organized mainly along a **molecular-anatomical gradient**. In mouse single-cell work, TRN neurons were described as spanning two negatively correlated transcriptional programs rather than breaking neatly into many unrelated clusters, with **Spp1** and **Ecel1** marking opposite poles of the gradient. Those poles also map onto differences in cell structure, physiology, thalamic targets, and sleep-related function. (Hartley et al., 2024) (Li et al., 2020)

At one end of this axis are **Spp1^+** neurons. These cells are located in the **central core** of the TRN, correspond to a **core/inner** anatomical organization, and are reported to have **high bursting activity**. Their efferents preferentially target **first-order thalamic nuclei**, and functional perturbation links this subnetwork to **sleep spindle generation** and increased **delta power** during slow-wave sleep. A later review also describes core/inner TRN neurons as having **larger somata** and **more extensive dendritic trees**, consistent with a higher capacity for synaptic integration, though this should be treated as a summary of prior work rather than a standalone primary classification. (Hartley et al., 2024) (Li et al., 2020) (Flores et al., 2026)

At the opposite end are **Ecel1^+** neurons. These cells lie in the **bordering shell** of the TRN, have a **shell-like/ edge** organization, and show **lower bursting activity** than Spp1^+ neurons. Their projections preferentially target **higher-order thalamic nuclei**, and their activity has the opposite sleep effect: they **antagonize sleep spindle duration** rather than promoting spindle expression. Together, these findings support the idea that internal TRN subtype structure is tied directly to distinct thalamocortical subnetworks rather than being only a molecular label. (Hartley et al., 2024) (Li et al., 2020)

A second, partly overlapping source of heterogeneity is the distinction between **PV-associated** and **non-PV** TRN neurons. One recent review states that **parvalbumin is expressed in about 70%** of TRN cells in mouse, implying that a substantial minority are not PV-positive. This fits with the broader picture already noted

above: PV is a very useful TRN marker in rodent anatomy, but it is not universal and should not be taken to define the whole nucleus by itself. (Flores et al., 2026) (Klimczak et al., 2024)

There is also **input-defined heterogeneity** across the thickness of sensory TRN. In somatosensory TRN, many neurons fit a tiered topography in which inner, middle, and outer positions receive input from different relay nuclei, but about **one quarter** of recorded neurons received **convergent input from multiple thalamic regions**, including sources usually located in different nuclei. These convergent cells were distributed across the thickness of TRN rather than confined to one lamina, suggesting that some TRN neurons are specialized for more integrative or cross-channel inhibitory control. (Lam et al., 2011)

Taken together, the best-supported subtype scheme from the supplied literature is therefore not a long list of fully discrete TRN classes, but a layered view of heterogeneity. One layer is the major **Spp1 core Ecel1 shell** transcriptomic and circuit axis; another is partial subdivision by markers such as **PV**; and another is local circuit specialization, including neurons with **single-source** versus **multi-source convergent** thalamic input. Across these schemes, the subtype features that are currently best characterized are anatomical position within TRN, burst propensity, thalamic target class, and role in sleep rhythms. (Hartley et al., 2024) (Li et al., 2020) (Lam et al., 2011)

Evidence

(Hartley et al., 2024)

Distinct structural and functional connectivity of genetically segregated thalamoreticular subnetworks

> *"The thalamic reticular nucleus (TRN), a thin shell of GABAergic neurons positioned between the thalamus and cortex, receives its excitatory drive from the collateral projections of both thalamocortical and corticothalamic neurons and sends its inhibitory input to the thalamus to modulate thalamic activity"*
>
> *".Our recent single-cell RNA sequencing analysis of the TRN demonstrated two neuronal populations defined by a transcriptomic gradient of over 100 differentially expressed genes. SPP1 and ECEL1 are the two genes most differentially expressed by the two transcriptional poles of this gradient. SPP1-expressing (Spp1+) and ECEL1-expressing (Ecel1+) neurons have distinct anatomical localization, efferent connectivity, electrophysiological profiles, and opposing functions in regulating sleep rhythms. (Li et al., 2020) Specifically, Spp1+ neurons are located in the central core of the TRN, exhibit high bursting activity, project to first-order (FO) thalamic nuclei, and contribute to sleep spindle generation and delta rhythm power during slow-wave sleep. Conversely, Ecel1+ neurons, found in the bordering shell of the TRN, exhibit low bursting activity, project to higher-order (HO) thalamic nuclei, and antagonize sleep spindle duration. In addition to distinct outputs to thalamic nuclei, the TRN also receives abundant inputs from cortical, thalamic, and other subcortical regions."*

(Li et al., 2020)

Distinct subnetworks of the thalamic reticular nucleus

> *"The thalamic reticular nucleus (TRN), the major source of thalamic inhibition, regulates thalamocortical interactions that are critical for sensory processing, attention and cognition1–5. TRN dysfunction has been linked to sensory abnormality, attention deficit and sleep disturbance across multiple neurodevelopmental disorders6–9. However, little is known about the organizational principles that underlie its divergent functions. Here we performed an integrative study linking single-cell molecular and electrophysiological features of the mouse TRN to connectivity and systems-level function. We found that cellular heterogeneity in the TRN is characterized by a transcriptomic gradient of two negatively correlated gene-expression profiles, each containing hundreds of genes. Neurons in the extremes of this transcriptomic gradient express mutually exclusive markers, exhibit core or shell-like anatomical structure and have distinct electrophysiological properties. The two TRN subpopulations make differential connections with the functionally distinct first-order and higher-order thalamic nuclei to form molecularly defined TRN–thalamus subnetworks. Selective perturbation of the two subnetworks in vivo revealed their differential role in regulating sleep. In sum, our study provides a comprehensive atlas of TRN neurons at single-cell resolution and links molecularly defined subnetworks to the functional organization of thalamocortical circuits. A study integrating single-cell RNA-sequencing and electrophysiology data shows that in mouse, the cellular repertoire of the thalamic reticular nucleus is characterized by a transcriptomic gradient defined at its extremes by mutually exclusive expression of Spp1 and Ecel1, providing insights into the organizational principles underlying the divergent functions of this brain region."*

(Flores et al., 2026)

Thalamic reticular nucleus in the pathophysiology of schizophrenia

> *"At the cellular level, the TRN is composed of GABAergic inhibitory neurons"*

"glutamate decarboxylase 67 and parvalbumin (PV), which is expressed in approximately 70% of these cells (Klimczak et al., 2024). These neurons exhibit distinct properties, including a specific topographic distribution. The TRN is primarily composed of two types of cells: core (inner neurons) and edge neurons, which have distinct structural and functional characteristics (Lam et al., 2011)(Li et al., 2020)(Martinez-Garcia et al., 2020). Core neurons are located mainly in the central region of the TRN and are distinguished by their larger somas and more extensive dendritic trees, allowing for greater integration of synaptic inputs. Additionally, these cells almost exclusively express secreted phosphoprotein 1 gene (Li et al., 2020)."

(Klimczak et al., 2024)

Long-term effects of a double hit murine model for schizophrenia on parvalbumin expressing cells and plasticity-related molecules in the thalamic reticular nucleus and the habenula

"Fast-spiking parvalbumin (PV) expressing GABAergic neurons play an important role in brain physiology by providing inhibitory control of cortical and subcortical circuits (Hu et al., 2014). They are involved in cortical maturation through activity-dependent changes and the control of critical periods (Carulli et al., 2021)(Reichelt et al., 2019). These inhibitory neurons undergo a protracted period of physiological maturation and integration into cortical and thalamic circuitry during early postnatal life (Reh et al., 2020)"

".In the thalamus, the TRN is the major source of GABAergic inhibition, and modulates the information between the cerebral cortex and the thalamic nuclei (Takata, 2019). Interestingly, many inhibitory neurons in this nucleus express PV and, in rodents, the thalamic PV+ inhibitory neurons are almost exclusively located in the TRN (Lein et al., 2007). The projections of these neurons contact with the rest of the thalamic nuclei, including the lateral posterior nucleus (LP), the homologous of the human pulvinar nucleus, and the mediodorsal nucleus (MD) (Pinault, 2004)."

(Lam et al., 2011)

Functional Organization of the Thalamic Input to the Thalamic Reticular Nucleus

"Most axons connecting the thalamus and cortex in both directions pass through the thalamic reticular nucleus (TRN), a thin layer of GABAergic cells adjacent to the thalamus, and innervate neurons there. The TRN, therefore, is in a strategic location to regulate thalamocortical communication. We recorded neurons of the somatosensory region of the TRN in a thalamocortical slice preparation and studied the spatial organization of their thalamic input using laser scanning photostimulation. We show that the thalamoreticular pathway is organized topographically for most neurons. The somatosensory region of the TRN can be organized into three tiers. From the inner (thalamoreticular) border to the outer, in a manner roughly reciprocal to the reticulothalamic pathway, each of these tiers receives its input from one of the somatosensory relays of the thalamus—the posterior medial, ventroposterior medial, and ventroposterior lateral nuclei. What is surprising is that approximately a quarter of the recorded neurons received input from multiple thalamic regions usually located in different nuclei. These neurons distribute evenly throughout the thickness of the TRN. Our results, therefore, suggest that there exist a subpopulation of TRN neurons that receive convergent inputs from multiple thalamic sources and engage in more complex patterns of inhibition of relay cells. We propose these neurons enable the TRN to act as an externally driven "searchlight" that integrates cortical and subcortical inputs and then inhibits or disinhibits specific thalamic relay cells, so that appropriate information can get through the thalamus to the cortex."